Analysis I

Uebungsblatt 7

Problem:	Comment:	Points:
1		28.5 / 31

1 (Convergence Radius I)

(a)

Series: $\displaystyle\sum_{n=0}^{\infty} \frac{z^n}{n^3+1} = \sum_{n=0}^{\infty} a_n z^n$, $a_n = \frac{1}{n^3+1}$.

Solution. Radius: $R = \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim \frac{n^3+1}{(n+1)^3+1} = 1$. - If $|z| < 1$, converges absolutely. - If $|z| > 1$, $\frac{|z|^n}{n^3+1} \not\to 0$, so diverges. - If $|z| = 1$, $\sum \frac{|z|^n}{n^3+1} \leq \sum \frac{1}{n^3}$, so absolute convergence.

Thus $R = 1$ and convergence for all $|z| \leq 1$. ✓ 3 □

(b)

Series: $\displaystyle\sum_{n=0}^{\infty} \frac{2^n}{3^{n^2}} z^n$, $a_n = \frac{2^n}{3^{n^2}}$.

Solution. $R = \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim \frac{1}{2}3^{2n+1} = +\infty$. The series converges for all $z \in \mathbb{C}$. □ ✓ 3

(c)

Series: $\displaystyle\sum_{n=0}^{\infty} \frac{n^5+1}{3^n+1} z^n$, $a_n = \frac{n^5+1}{3^n+1}$.

Solution. $R = \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim \frac{n^5+1}{(n+1)^5+1} \cdot \frac{3^{n+1}+1}{3^n+1} = 3$. ✓ - If $|z| < 3$, absolute convergence. - If $|z| > 3$, divergence. - If $|z| = 3$, $|a_n z^n| = (n^5+1)\frac{3^n}{3^n+1} \sim n^5 \not\to 0$, ✓ so divergence.

Hence $R = 3$ and divergence on $|z| = 3$. 3 □

2 (Convergence Radius II)

Let $\sum_{n=0}^{\infty} a_n x^n$ have radius 1.

(a)

Let $0 < B < 1$. Show: for every $\varepsilon > 0$ there exists $N = N(\varepsilon, B)$ such that $\sum_{n=N}^{\infty} |a_n||y|^n \leq \varepsilon$ for all $|y| < B$.

Solution. Pick r with $B < r < 1$ and set $\rho := B/r < 1$. Since $\sum |a_n| r^n$ converges, its tails tend to 0. For $|y| < B$ and $n \geq N$, $\left(\frac{|y|}{r}\right)^n \leq \rho^n \leq \rho^N$, hence

$$\sum_{n=N}^{\infty} |a_n||y|^n = \sum_{n=N}^{\infty} |a_n| r^n \left(\frac{|y|}{r}\right)^n \leq \rho^N \sum_{n=N}^{\infty} |a_n| r^n.$$

The right-hand side tends to 0 as $N \to \infty$, so choose N with $\rho^N \sum_{n=N}^{\infty} |a_n| r^n \leq \varepsilon$. This N works for all $|y| < B$. $\square$

3

(b)

Let $(x_m)_m$ be a sequence with $|x_m| < 1$ for all $m \in \mathbb{N}$ and suppose $x_m \to x$ for some $x \in \mathbb{R}$ with $|x| < 1$. Show that there is a $B < 1$ such that

$$\sup\{|x_m|, |x|\} \leq B.$$

Solution. Pick a buffer $\varepsilon := \frac{1-|x|}{2} > 0$, so that $|x| + \varepsilon = \frac{|x|+1}{2} =: B_{\text{tail}} < 1$. By convergence, there exists $N \in \mathbb{N}$ such that for all $m \geq N$ we have $|x_m - x| < \varepsilon$. Hence, for all $m \geq N$,

$$|x_m| \leq |x| + |x_m - x| < |x| + \varepsilon = B_{\text{tail}} < 1.$$

Define

$$B := \max\big\{B_{\text{tail}}, |x_1|, |x_2|, \ldots, |x_{N-1}|\big\}.$$

Each quantity inside the maximum is strictly smaller than 1 (by hypothesis and because $B_{\text{tail}} < 1$), hence $B < 1$. For $m \geq N$ we have $|x_m| \leq B_{\text{tail}} \leq B$, while for $1 \leq m < N$ we have $|x_m| \leq B$ by construction; clearly also $|x| \leq B_{\text{tail}} \leq B$. Therefore

$$\sup\{|x_m|, |x|\} \leq B < 1.$$

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$\square$

(c)

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence 1. Suppose $(x_m)_{m\in\mathbb{N}}$ satisfies $|x_m| < 1$ for all m and $x_m \to x$ with $|x| < 1$. We show

$$\lim_{m\to\infty} \sum_{n=0}^{\infty} a_n x_m^n \;=\; \sum_{n=0}^{\infty} a_n x^n.$$

Solution. By part (b), there exists $B < 1$ such that $\sup\{|x_m|, |x|\} \leq B$. By part (a), for every $\varepsilon > 0$ there is $N = N(\varepsilon, B) \in \mathbb{N}$ such that for all $|y| \leq B$,

$$\sum_{n>N} |a_n|\, |y|^n \leq \varepsilon.$$

Fix $\varepsilon > 0$ and choose N accordingly. For any $m \in \mathbb{N}$ we write

$$\sum_{n=0}^{\infty} a_n x_m^n - \sum_{n=0}^{\infty} a_n x^n = \underbrace{\sum_{n=0}^{N} a_n \left(x_m^n - x^n\right)}_{\text{finite part}} + \underbrace{\sum_{n>N}^{\infty} a_n x_m^n}_{\text{tail at } x_m} - \underbrace{\sum_{n>N}^{\infty} a_n x^n}_{\text{tail at } x}.$$

Taking absolute values and applying the triangle inequality gives

$$\Big|\sum_{n=0}^{\infty} a_n x_m^n - \sum_{n=0}^{\infty} a_n x^n\Big| \;\leq\; \Big|\sum_{n=0}^{N} a_n \left(x_m^n - x^n\right)\Big| \;+\; \sum_{n>N}^{\infty} |a_n|\, |x_m|^n \;+\; \sum_{n>N}^{\infty} |a_n|\, |x|^n.$$

Since $|x_m| \le B$ and $|x| \le B$, the last two terms are bounded by ε each by (a). For the finite part, use the algebraic identity $a^n - b^n = (a-b)\sum_{k=0}^{n-1} a^{n-1-k}b^k$ to get, for $n \ge 1$,

$$|x_m^n - x^n| \le |x_m - x| \sum_{k=0}^{n-1} |x_m|^{n-1-k}|x|^k \le |x_m - x|\, nB^{n-1}.$$

Therefore

$$\Big|\sum_{n=0}^{N} a_n\,(x_m^n - x^n)\Big| \le |x_m - x| \sum_{n=1}^{N} n|a_n|B^{n-1} =: C_{N,B}\,|x_m - x|.$$

Because $x_m \to x$, there exists m_ε such that for all $m \ge m_\varepsilon$ we have $|x_m - x| \le \varepsilon / C_{N,B}$, and hence

$$\Big|\sum_{n=0}^{\infty} a_n x_m^n - \sum_{n=0}^{\infty} a_n x^n\Big| \le C_{N,B}|x_m - x| + \varepsilon + \varepsilon \le 3\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, the claim follows. □

3 (Normed Vector Spaces I)

Solution. Example in $\mathbb{R}^2$: take $x = (4,0)$ and $y = (3,3)$. Then $\|x\|_2 = 4$, $\|y\|_2 = \sqrt{18} > 4$, while $\|x\|_\infty = 4$ and $\|y\|_\infty = 3 < 4$.

Technically n arbitrary

This does not contradict norm equivalence: in finite dimensions there exist constants $C_1, C_2 > 0$ with $C_1\|v\|_\infty \le \|v\|_2 \le C_2\|v\|_\infty$ for all v, but equivalence does not preserve the ordering between *different* vectors. □

Instead, norm equivalence means that the norms are "comparable" for any *single* vector. Formally, it means there exist positive constants C_1 and C_2 such that for **any vector** v, the following inequality holds:

$$C_1\|v\|_\infty \le \|v\|_2 \le C_2\|v\|_\infty$$

(For $\mathbb{R}^n$, these constants are $C_1 = 1$ and $C_2 = \sqrt{n}$.)

5.5

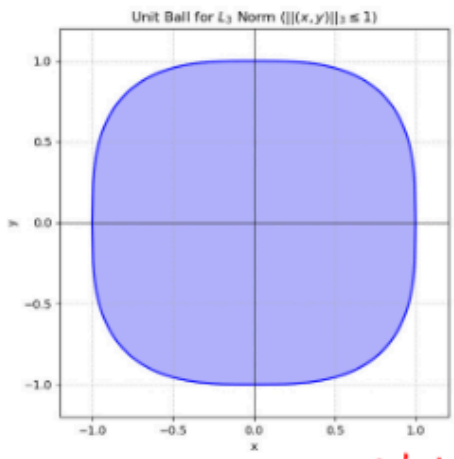
4 (Normed Vector Spaces II)

For $x, y \in \mathbb{R}$, we write (x,y) for the vector in $\mathbb{R}^2$ with components x and y.

For $x, y \in \mathbb{R}$, write (x,y) for the vector in $\mathbb{R}^2$.

(a)

$B(0,1) = \{(x,y) : |x|^3 + |y|^3 \le 1\}$; its boundary is $|x|^3 + |y|^3 = 1$ (symmetric "supercircle" with intercepts $(\pm 1, 0)$ and $(0, \pm 1)$).



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You need to prove this!

(b)

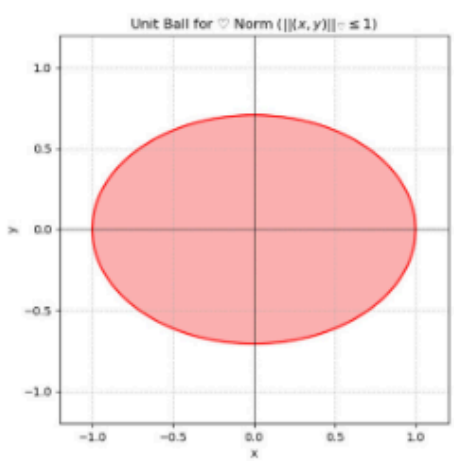
Solution. ~~Clearly~~ $\|v\|_\heartsuit \geq 0$ and $\|v\|_\heartsuit = 0 \iff v = 0$. For $\alpha \in \mathbb{R}$, $\|\alpha v\|_\heartsuit = \sqrt{(\alpha x)^2 + 2(\alpha y)^2} = |\alpha|\,\|v\|_\heartsuit$. For the triangle inequality define $T(x,y) = (x, \sqrt{2}\,y)$. Then $\|v\|_\heartsuit = \|T(v)\|_2$, hence $\|v+w\|_\heartsuit = \|T(v+w)\|_2 \leq \|T(v)\|_2 + \|T(w)\|_2 = \|v\|_\heartsuit + \|w\|_\heartsuit$. $\square$

You need to prove that T is linear for this.

2

(c)

$B(0,1) = \{(x,y) : x^2 + 2y^2 \leq 1\}$, an ellipse with axes intercepts $(\pm 1, 0)$ and $(0, \pm 1/\sqrt{2})$.



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